```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style='whitegrid')

df = sns.load_dataset('healthexp')
print("Shape:", df.shape)
print(df.head())
print("\nMissing values per column:")
print(df.isnull().sum())
print(df.dtypes)
```

```
Shape: (274, 4)
   Year        Country  Spending_USD  Life_Expectancy
0  1970        Germany       252.311             70.6
1  1970         France       192.143             72.2
2  1970  Great Britain       123.993             71.9
3  1970          Japan       150.437             72.0
4  1970            USA       326.961             70.9

Missing values per column:
Year               0
Country            0
Spending_USD       0
Life_Expectancy    0
dtype: int64
Year                int64
Country            object
Spending_USD      float64
Life_Expectancy   float64
dtype: object
```

```
print("Countries included:")
print(df["Country"].unique())
print(f"\nYear range: {df['Year'].min()} to {df['Year'].max()}\n")
print("Summary Statistics:")
display(df.describe())
```

```
Countries included:
['Germany' 'France' 'Great Britain' 'Japan' 'USA' 'Canada']

Year range: 1970 to 2020

Summary Statistics:
```

	Year	Spending_USD	Life_Expectancy
count	274.000000	274.000000	274.000000
mean	1996.992701	2789.338905	77.909489
std	14.180933	2194.939785	3.276263
min	1970.000000	123.993000	70.600000
25%	1985.250000	1038.357000	75.525000
50%	1998.000000	2295.578000	78.100000
75%	2009.000000	4055.610000	80.575000
max	2020.000000	11859.179000	84.700000

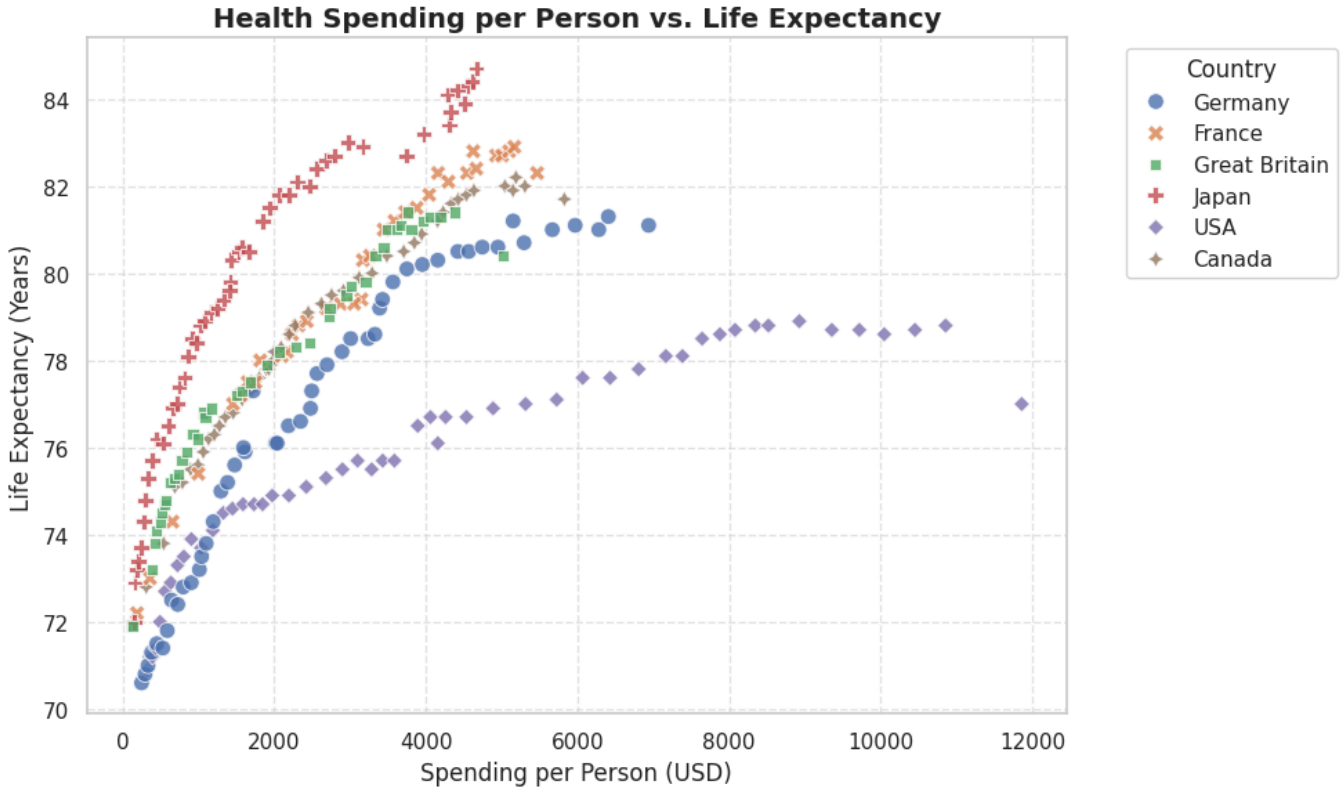
Task 1 — The Core Relationship

```
plt.figure(figsize=(10, 6))

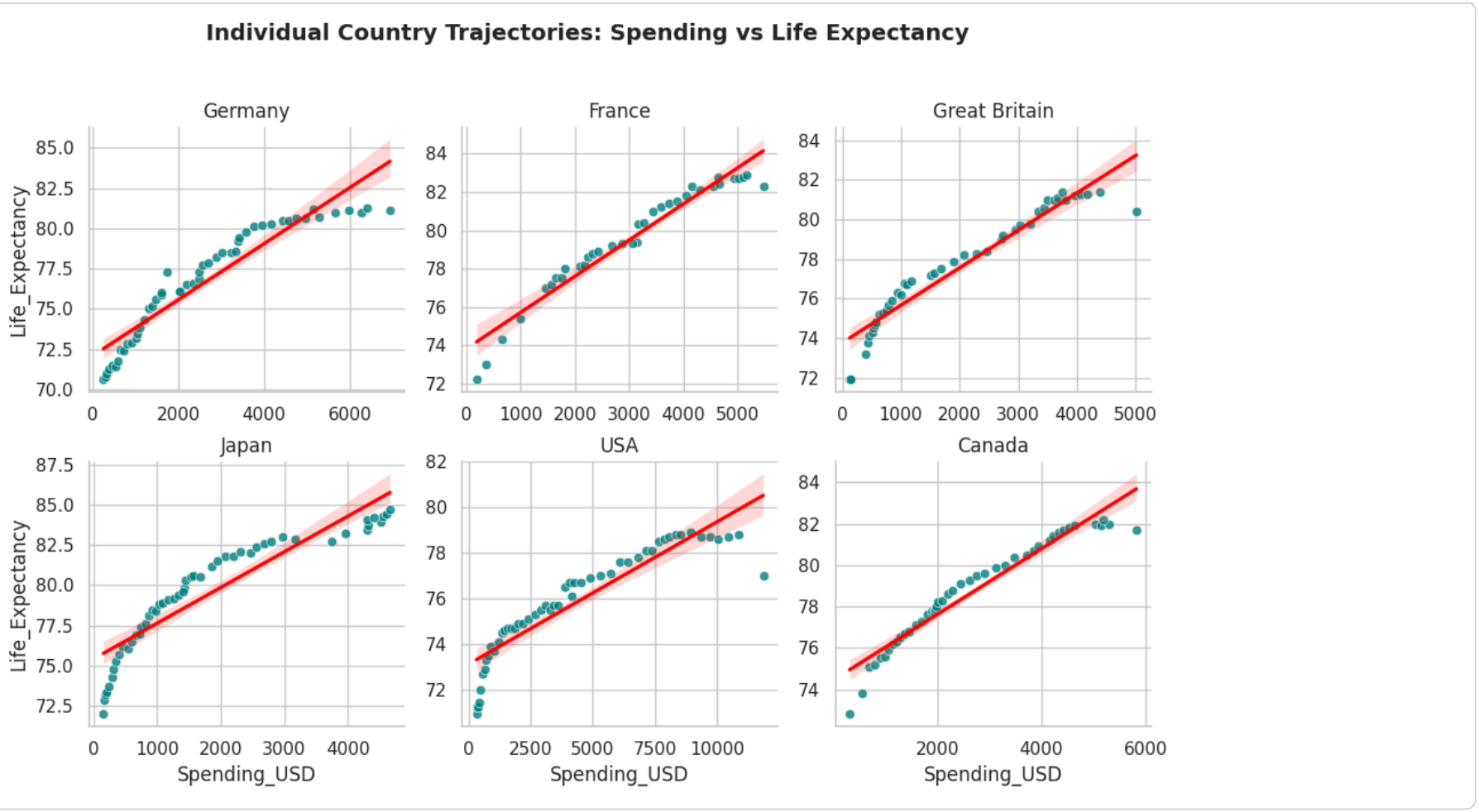
sns.scatterplot(
    data=df,
    x='Spending_USD',
    y='Life_Expectancy',
    hue='Country',
    style='Country',
    s=70,
    alpha=0.8
)

plt.title('Health Spending per Person vs. Life Expectancy', fontsize=14, fontweight='bold')
plt.xlabel('Spending per Person (USD)', fontsize=12)
plt.ylabel('Life Expectancy (Years)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend(title='Country', bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()
plt.show()
```




```
# Individual country relationships
g = sns.FacetGrid(df, col='Country', col_wrap=3, height=3.5, sharex=False, sharey=False)
g.map_dataframe(sns.scatterplot, x='Spending_USD', y='Life_Expectancy', color='teal', alpha=0.8)
g.map_dataframe(sns.regplot, x='Spending_USD', y='Life_Expectancy', scatter=False, color='red')
g.set_titles(col_template="{col_name}")
g.fig.subplots_adjust(top=0.85)
g.fig.suptitle('Individual Country Trajectories: Spending vs Life Expectancy', fontsize=14, fontweight='bold')
plt.show()
```



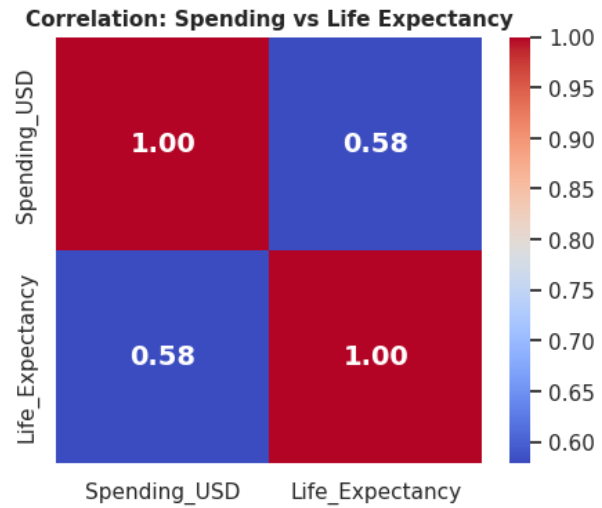
✍ **Q1:** In general, as spending goes up, does life expectancy go up, down, or stay flat? Would you call the relationship strong or weak?
Optional hint for evidence: df['Spending_USD'].corr(df['Life_Expectancy']) gives a number from −1 to +1.

As spending goes up, life expectancy goes UP. Positive (+0.58), confirming that life expectancy rises as health spending increases.

```
corr_matrix = df[['Spending_USD', 'Life_Expectancy']].corr()

# Single Heatmap Plot
plt.figure(figsize=(5, 4))
sns.heatmap(
    corr_matrix,
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    cbar=True,
    square=True,
```

```
    annot_kws={'size': 14, 'weight': 'bold'},
)
plt.title(
    'Correlation: Spending vs Life Expectancy', fontsize=11, fontweight='bold'
)
plt.show()
```



Task 2 — Find the Odd One Out

```
# 1. Filter data for the most recent year
latest = df[df['Year'] == df['Year'].max()]

# 2. Sort by Spending_USD to compare spending vs life expectancy
latest_sorted = latest.sort_values(by='Spending_USD', ascending=False)
print("--- Latest Year Data (Sorted by Spending) ---")
display(latest_sorted[['Country', 'Spending_USD', 'Life_Expectancy']])

# 3. Bar Chart: Health Spending per Country (Latest Year)
plt.figure(figsize=(10, 5))
sns.barplot(
    data=latest_sorted,
    x='Country',
    y='Spending_USD',
    palette='Reds_r',
    hue='Country',
    legend=False,
)
plt.title(
    f'Health Spending per Country in {df["Year"].max()}',
    fontsize=13,
    fontweight='bold',
)
plt.xlabel('Country')
plt.ylabel('Spending per Person (USD)')
```

```
plt.grid(True, axis='y', linestyle='--', alpha=0.5)

# Add data labels on top of bars
for p in plt.gca().patches:
  plt.gca().annotate(
      f'${p.get_height():,.0f}',
      (p.get_x() + p.get_width() / 2.0, p.get_height()),
      ha='center',
      va='center',
      xytext=(0, 5),
      textcoords='offset points',
  )

plt.show()

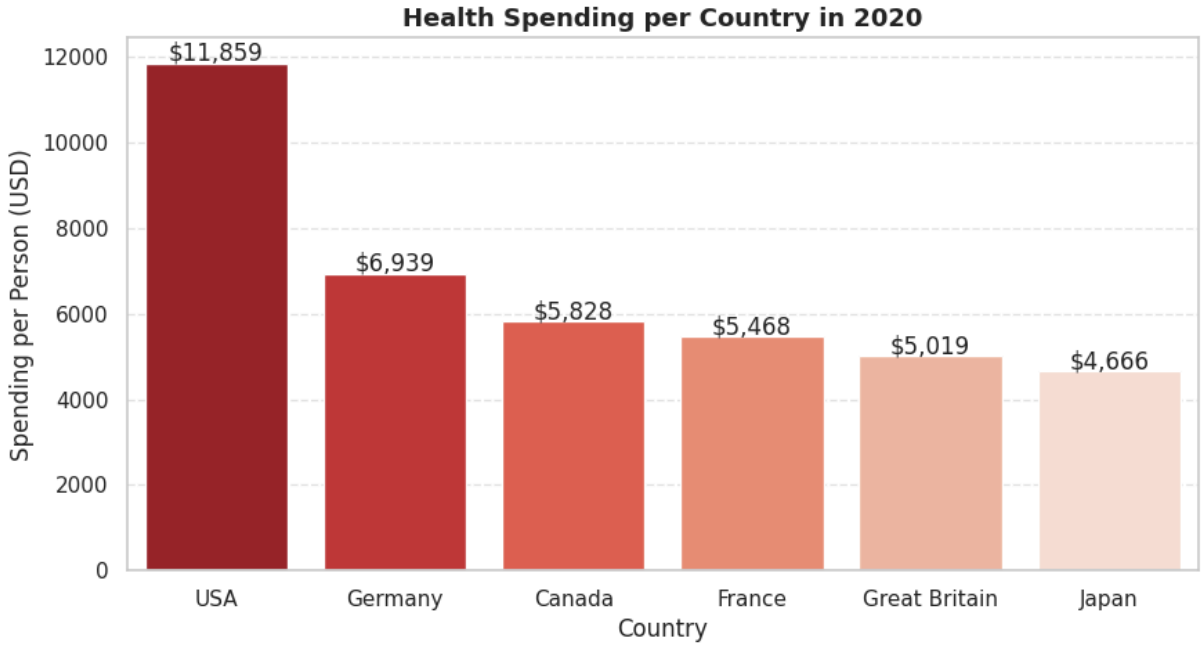
# 4. Complementary Bar Chart: Life Expectancy per Country (Latest Year)
plt.figure(figsize=(10, 5))
latest_life_sorted = latest.sort_values(
    by='Life_Expectancy', ascending=False
)
sns.barplot(
    data=latest_life_sorted,
    x='Country',
    y='Life_Expectancy',
    palette='Greens_r',
    hue='Country',
    legend=False,
)
plt.title(
    f'Life Expectancy per Country in {df["Year"].max()}',
    fontsize=13,
    fontweight='bold',
)
plt.xlabel('Country')
plt.ylabel('Life Expectancy (Years)')
plt.ylim(75, 86)  # Zoom in to highlight variations
plt.grid(True, axis='y', linestyle='--', alpha=0.5)

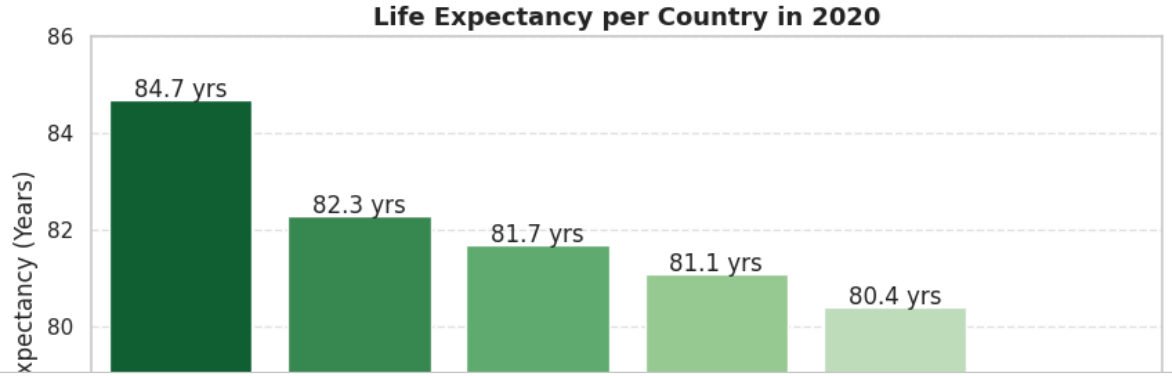
for p in plt.gca().patches:
  plt.gca().annotate(
      f'{p.get_height():.1f} yrs',
      (p.get_x() + p.get_width() / 2.0, p.get_height()),
      ha='center',
      va='center',
      xytext=(0, 5),
      textcoords='offset points',
  )

plt.show()
```

```
--- Latest Year Data (Sorted by Spending) ---
```

	Country	Spending_USD	Life_Expectancy
273	USA	11859.179	77.0
269	Germany	6938.983	81.1
268	Canada	5828.324	81.7
270	France	5468.418	82.3
271	Great Britain	5018.700	80.4
272	Japan	4665.641	84.7



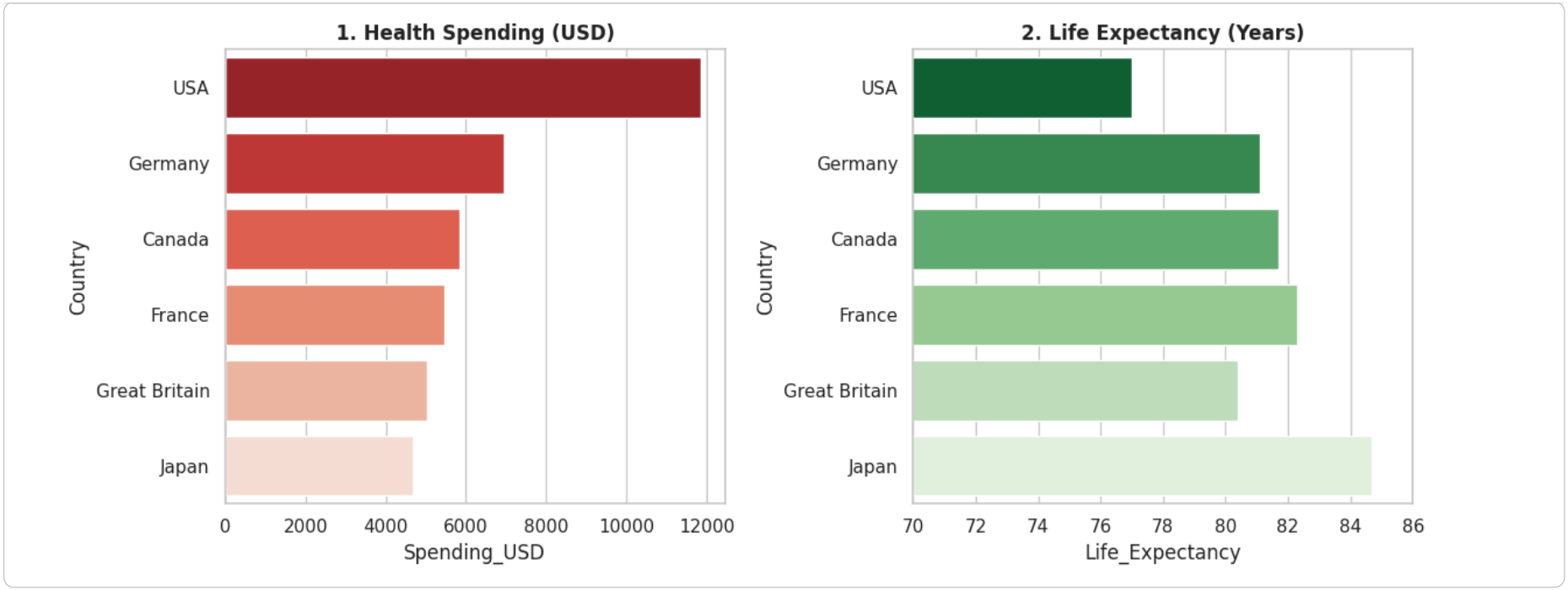



```
# Subplot: Spending vs Life Expectancy Comparison
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

```
# Plot 1: Spending
sns.barplot(
    data=latest_sorted,
    x='Spending_USD',
    y='Country',
    ax=axes[0],
    palette='Reds_r',
    hue='Country',
    legend=False,
)
axes[0].set_title(
    '1. Health Spending (USD)', fontsize=12, fontweight='bold'
)

# Plot 2: Life Expectancy
sns.barplot(
    data=latest_sorted,
    x='Life_Expectancy',
    y='Country',
    ax=axes[1],
    palette='Greens_r',
    hue='Country',
    legend=False,
)
axes[1].set_title(
    '2. Life Expectancy (Years)', fontsize=12, fontweight='bold'
)
axes[1].set_xlim(70, 86)  # Zoom in for comparison

plt.tight_layout()
plt.show()
```



✍ **Q2:** Which country spends the most per person? Which country lives the longest? Are they the same country? (This is the heart of the assignment — what surprised you?) Your answer: _____

Spends the most: USA (Spends significantly more than any other nation at $11,859 per person).

Lives the longest: Japan (Achieves the highest life expectancy at 84.7 years).

Are they the same country? No, they are completely different.

Surprising Analytical Finding (Analytical Narrative): What surprises me most is the stark efficiency gap in healthcare spending. Despite the USA spending more than double the per-person healthcare budget of Japan ($11,859 vs. 4,666$), it ranks last in life expectancy (77.0 years) among the six analyzed nations. Conversely, Japan achieves the highest lifespan while spending the least per capita. This reveals that higher overall expenditure does not automatically translate to better healthcare outcomes or longevity.

Task 3 — Change Over Time

```
sns.set_theme(style='whitegrid')

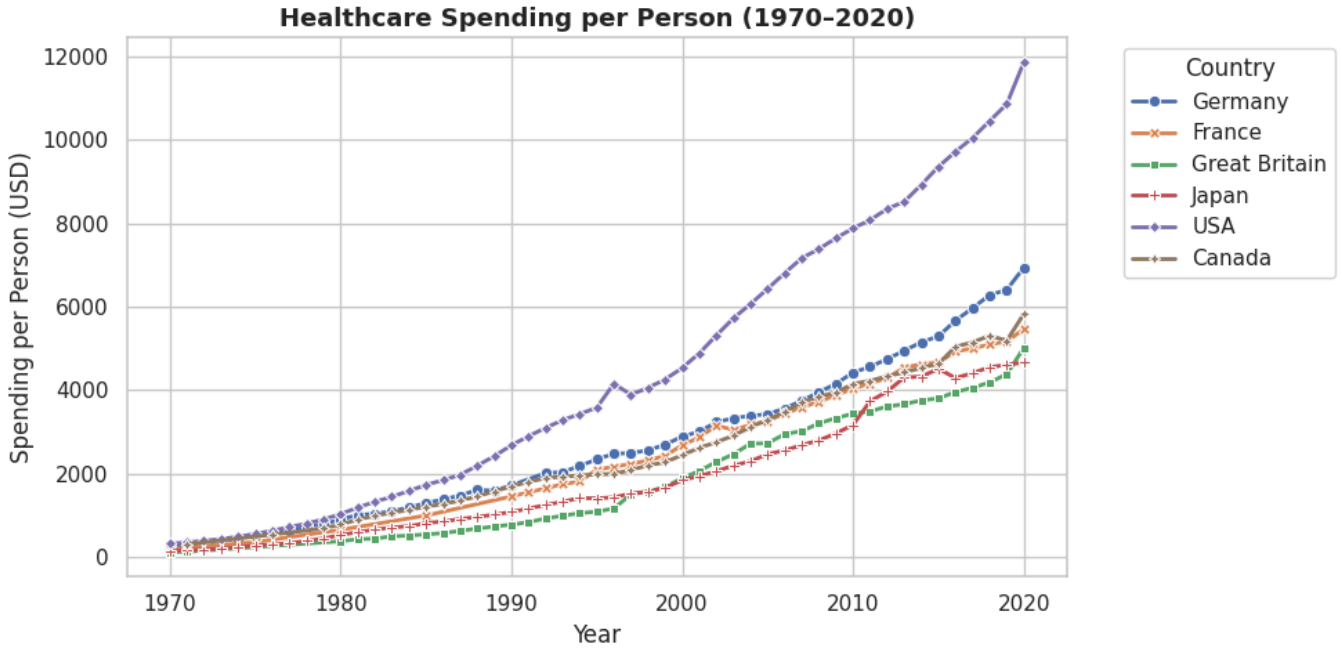
# ------------------------------------------------------------
# Option A: Primary Solution (Spending_USD over Time)
# ------------------------------------------------------------
plt.figure(figsize=(10, 5))
sns.lineplot(
    data=df,
    x='Year',
```

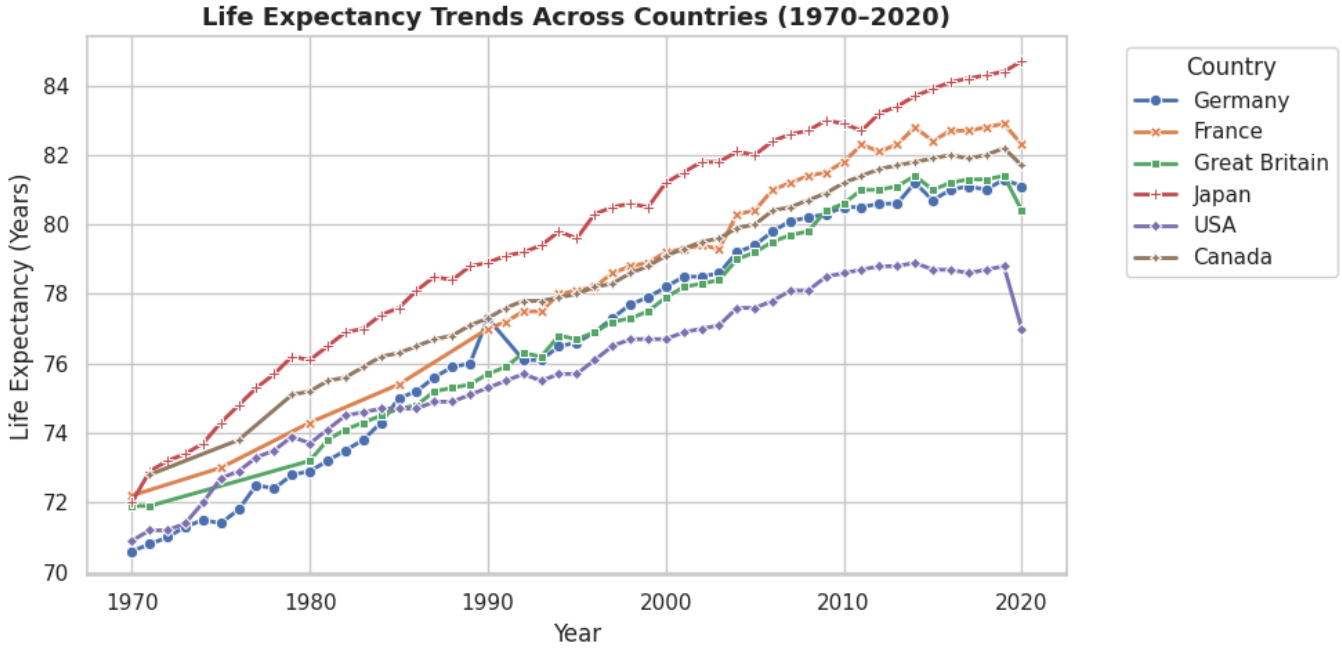
```
    y='Spending_USD',
    hue='Country',
    style='Country',
    markers=True,
    dashes=False,
    linewidth=2,
)

plt.title(
    'Healthcare Spending per Person (1970–2020)',
    fontsize=13,
    fontweight='bold',
)
plt.xlabel('Year')
plt.ylabel('Spending per Person (USD)')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', title='Country')
plt.tight_layout()
plt.show()

# ------------------------------------------------------------
# Extension Solution: Life Expectancy Trajectory over Time
# ------------------------------------------------------------
plt.figure(figsize=(10, 5))
sns.lineplot(
    data=df,
    x='Year',
    y='Life_Expectancy',
    hue='Country',
    style='Country',
    markers=True,
    dashes=False,
    linewidth=2,
)

plt.title(
    'Life Expectancy Trends Across Countries (1970–2020)',
    fontsize=13,
    fontweight='bold',
)
plt.xlabel('Year')
plt.ylabel('Life Expectancy (Years)')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', title='Country')
plt.tight_layout()
plt.show()
```

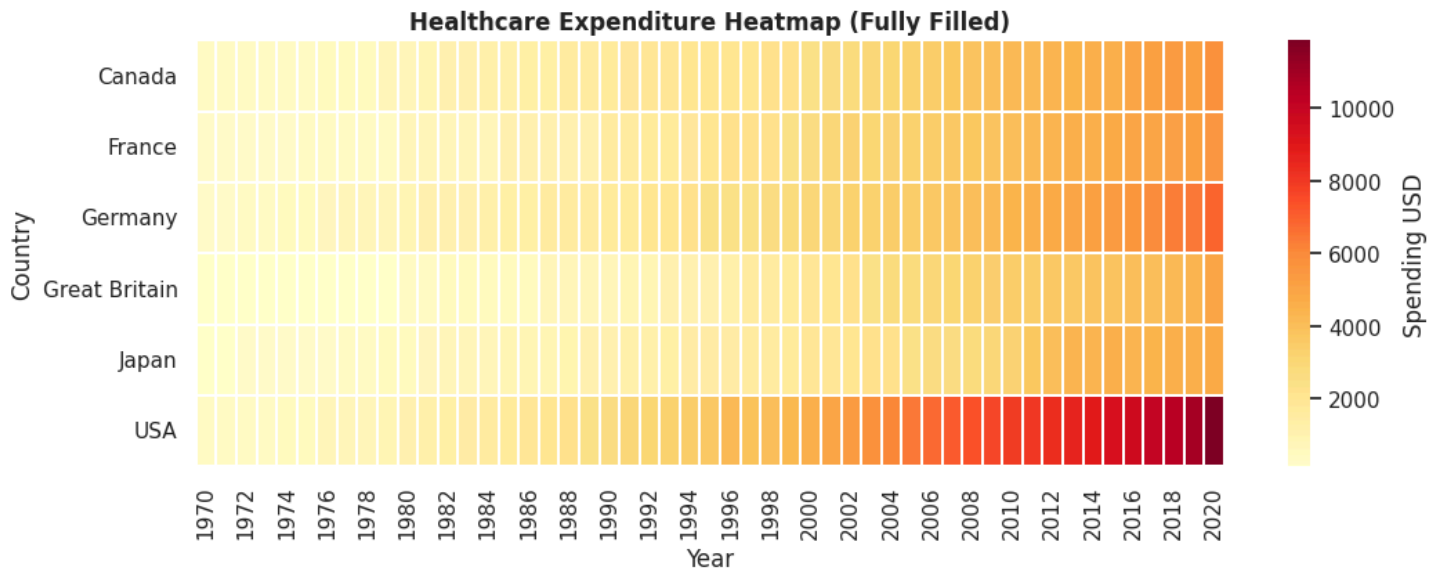





```
# Forward fill followed by backward fill to handle initial missing years
pivot_spending = (
    df.pivot(index='Country', columns='Year', values='Spending_USD')
```

```
    .ffill(axis=1)
    .bfill(axis=1)
)

plt.figure(figsize=(12, 4))
sns.heatmap(
    pivot_spending,
    cmap='YlOrRd',
    cbar_kws={'label': 'Spending USD'},
    linewidths=0.1,
)
plt.title(
    'Healthcare Expenditure Heatmap (Fully Filled)',
    fontsize=12,
    fontweight='bold',
)
plt.xlabel('Year')
plt.ylabel('Country')
plt.show()
```



✍ Q3: Describe one clear trend you see over time. Does one country pull away from the others at some point? Your answer:

_____

Clear Trend Over Time: Across all six nations, both healthcare spending per person and life expectancy have steadily increased from 1970 to 2020.

Country Pulling Away: Yes, the USA sharply pulls away in healthcare expenditure.

Task 4 — Your Own Investigation

```
# Question: How did each country shift in Life Expectancy from 1970 to 2020?

# Data prep
df_1970 = df[df['Year'] == 1970][['Country', 'Life_Expectancy']].rename(
    columns={'Life_Expectancy': 'LE_1970'}
)
df_2020 = df[df['Year'] == df['Year'].max()][
    ['Country', 'Life_Expectancy']
].rename(columns={'Life_Expectancy': 'LE_2020'})
merged = pd.merge(df_1970, df_2020, on='Country').sort_values(
    by='LE_2020', ascending=True
)

plt.figure(figsize=(10, 5))

# Connecting lines
plt.hlines(
    y=merged['Country'],
    xmin=merged['LE_1970'],
    xmax=merged['LE_2020'],
    color='gray',
    alpha=0.6,
    linewidth=2,
)

# Dots for 1970 and 2020
plt.scatter(
    merged['LE_1970'],
    merged['Country'],
    color='skyblue',
    alpha=1,
    s=100,
    label='1970 Baseline',
```